

## Homework 3: What the Judges Actually Score

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*Paper track. Original scenario, invitational format. Do this on your own.*

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### The Scene

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At a weekend robotics tournament, the club's cash bag (about \$600 in registration fees) disappears from the check-in table between 8:15 and 8:40 a.m. The table stands on a strip of exposed dirt just inside the gym's side door.

Investigators recover: a smear of **white powder** on the table edge that **turns iodine blue-black** and does **not** fizz with acid; two **short white animal hairs** on the cash bag strap; a **synthetic (polyester) fiber**, navy blue, caught on the zipper; a small crumb of **dark, fine, iron-rich reddish soil**; and a **partial fingerprint (tented arch)** on the metal clasp.

### The Suspects

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- **Reese** — runs the snack table (which has flour, powdered sugar, and corn starch out for a baking demo). Wears a navy cotton club shirt. Has a white cat. Was refilling the coffee urn at 8:20 (two witnesses). Fingerprints: ulnar loop, ulnar loop.
- **Sky** — the tournament's setup volunteer; had a key to the cash bag lock. Wears a navy **polyester** windbreaker. No pets. "Checking the field alone" from 8:10 to 8:45. Fingerprints: tented arch, radial loop.
- **Devon** — maintains the school garden; keeps a bin of **reddish iron-rich potting soil** by the gym's side door. Wears a gray fleece. Has a white dog. Fingerprints: plain whorl, ulnar loop.
- **Bailey** — was denied a refund last week and complained loudly to the club treasurer. Runs the 3-D-printing booth (lots of PLA plastic filament, no powders). Wears a green wool sweater. No pets. Fingerprints: plain arch, tented arch.

### Problems

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**1. (5 pts)** Complete the evidence grid. For each item: which suspect(s) does it point to, and is it a **strong** (unusual/specific) or **common** match?

| Evidence                                | Points to | Strong or common? |
|-----------------------------------------|-----------|-------------------|
| White powder — iodine-positive, no fizz |           |                   |
| Short white animal hair                 |           |                   |
| Navy polyester fiber                    |           |                   |
| Reddish iron-rich soil                  |           |                   |
| Tented arch fingerprint                 |           |                   |

**2. (2 pts)** Which suspect has the strongest **motive**? Do the physical evidence and that suspect line up? What does that tell you?

**3. (10 pts)** Write a full analysis: name the culprit; give a motive; list the evidence against them, each with a reason; give a timeline note; and clear each of the other three suspects in a sentence.

**4. (1 pt)** The white powder is iodine-positive and doesn't fizz. Name a powder it could be, with its formula.

**5. (2 pts)** Explain why "the tented arch fingerprint matches Sky and Bailey" is *class* evidence, not proof.

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